Individual Contribution Report

Operating Systems Final Project · MiniOS Ekoms

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Group: 8
Assigned Role: Process and Memory Management

What did I contribute to the project?
I built the process and memory management modules (processes.py and memory.py) and integrated them with the shell commands in main.py.

What features did I personally develop?
I created a process manager (with a PCB class, automatic PIDs, a 20-process limit, and control commands) and a memory manager (modeling 1024 MB total memory, allocation commands, and a visual progress bar).

What problems did I encounter?
I faced crashing user inputs, memory over-allocation, built-in exception naming conflicts, invalid PID/table limits, scaling UI elements, and aligning error handling with the shared shell.

How did I solve those problems?
I added try/except input validation, enforced strict memory and process limits, renamed my custom exception to MemoryError_, and configured the shell to safely catch errors.

What did I learn from this project?
I learned core OS concepts like process tables and memory partitioning, the critical need for input validation, and how early team communication ensures smooth module integration.

Evidence of my contribution

GitHub repository: GitHub repository: https://github.com/OtitsDub/Group8-SC1063

Source code: memory.py
processes.py

Documentation: README.md

Screenshots:



